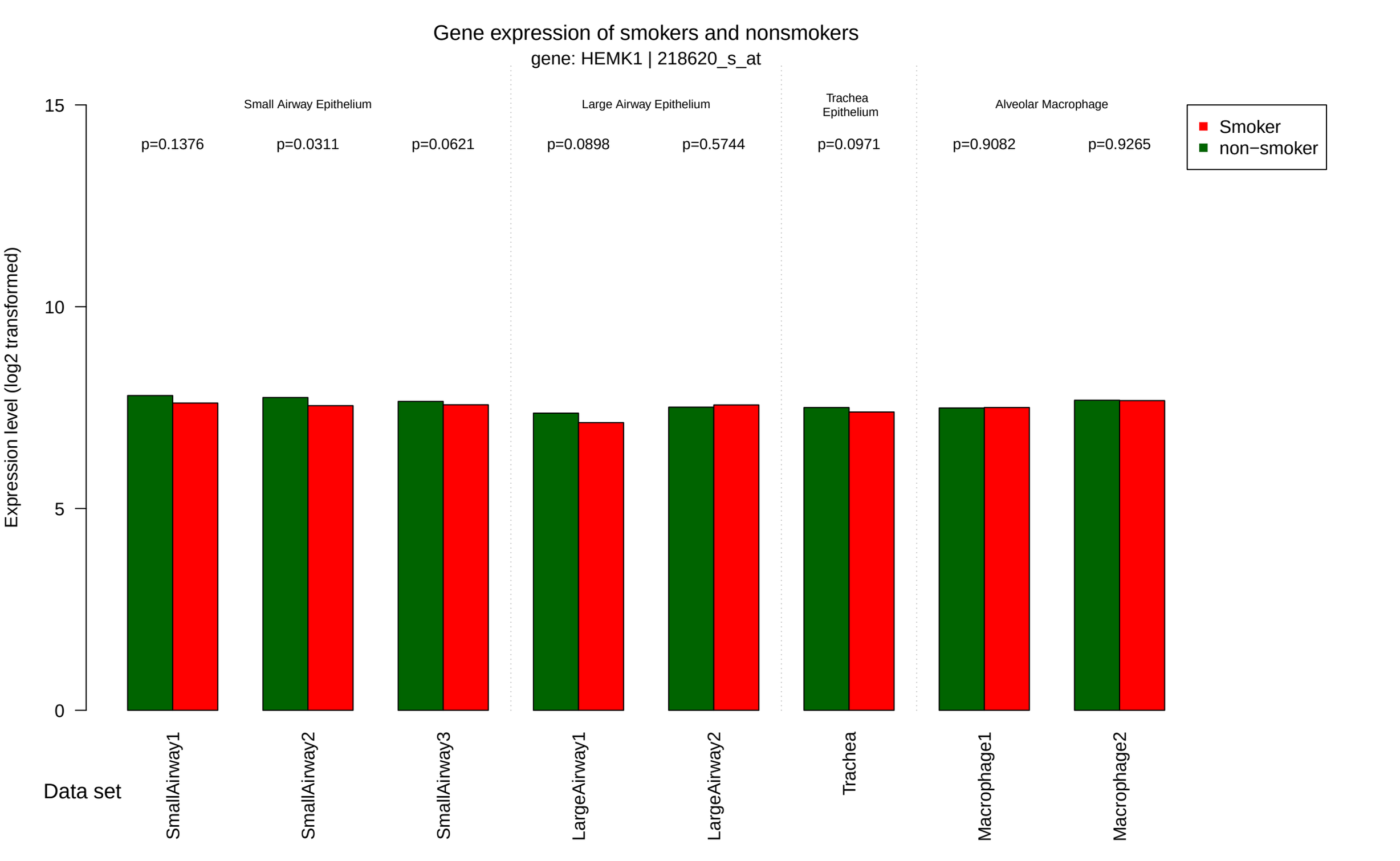
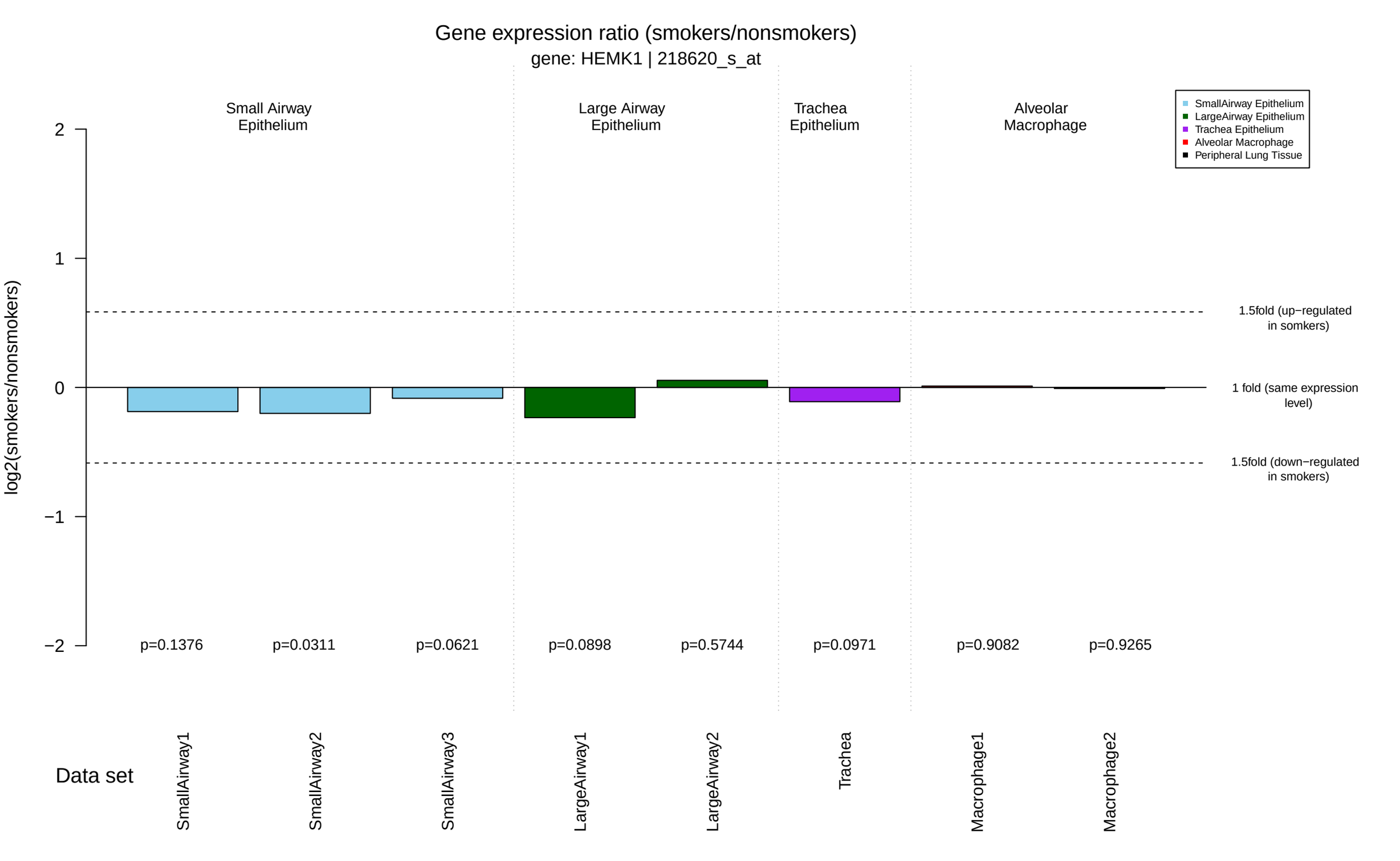
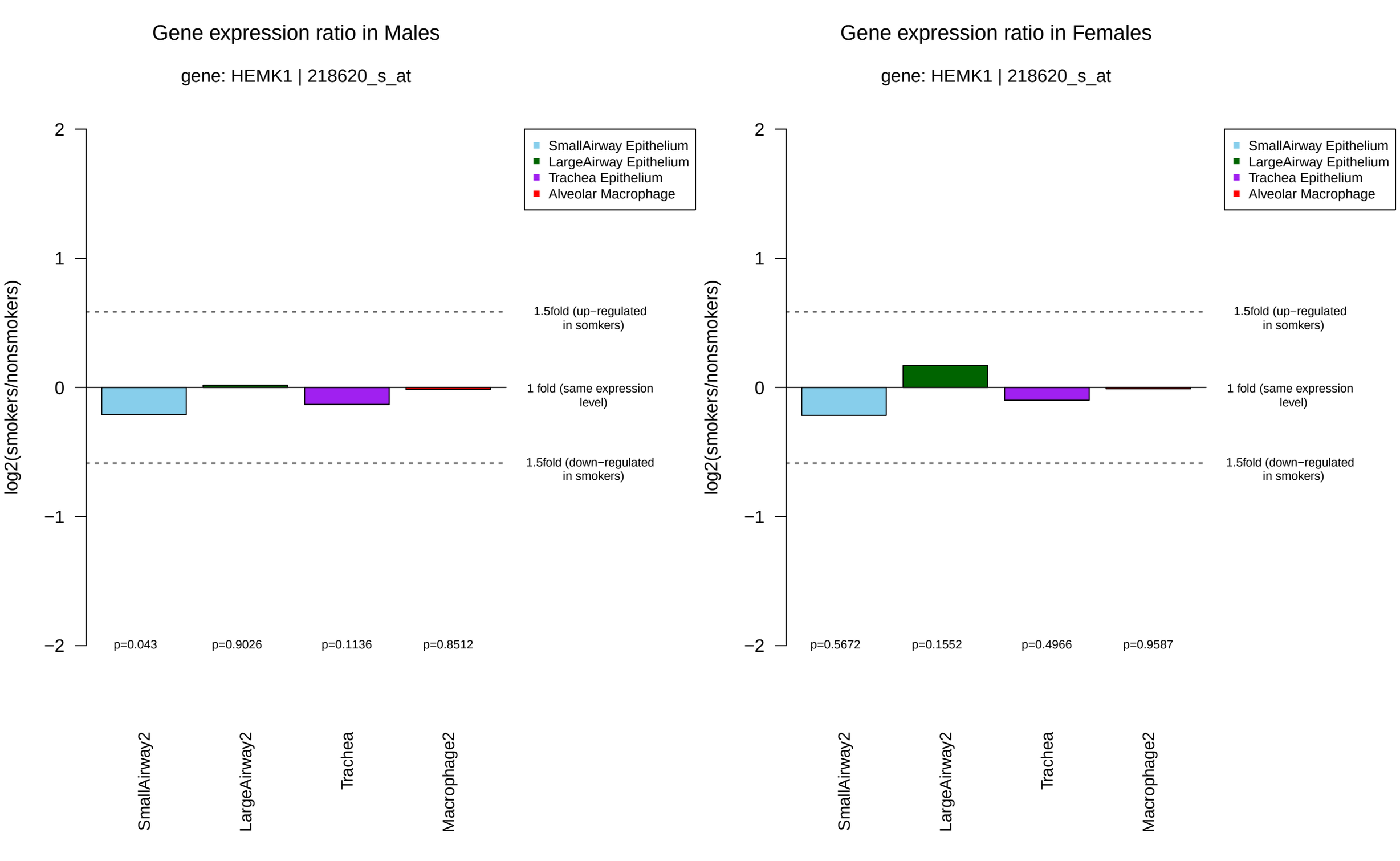
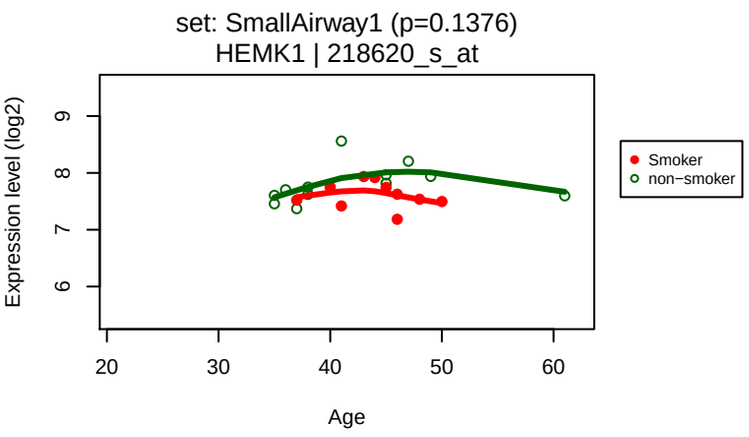
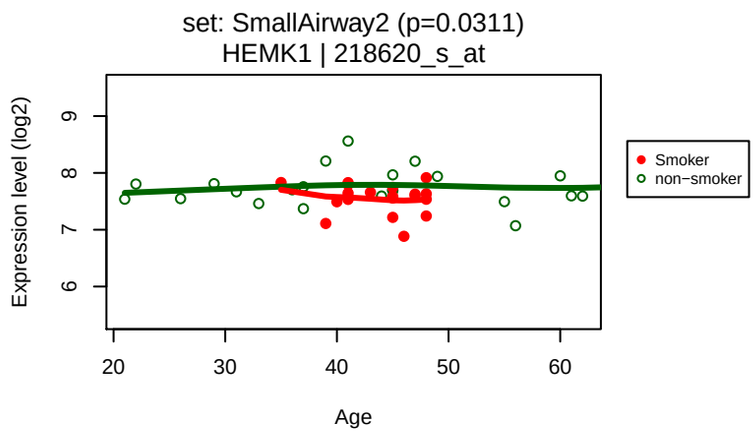
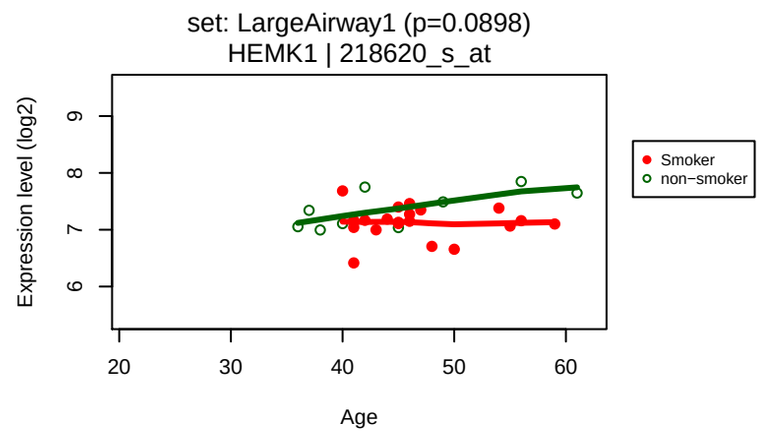

Gene expression of smokers and nonsmokers
gene: HEMK1 | 218620_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1376
p=0.0311
p=0.0621
p=0.0898
p=0.5744
p=0.0971
p=0.9082
p=0.9265
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

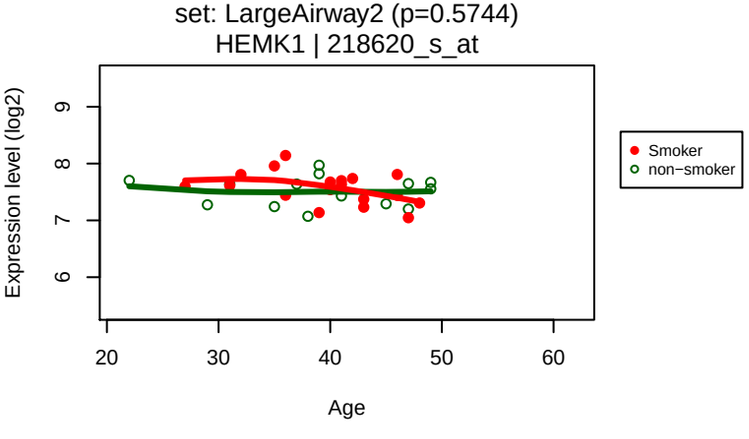

Gene expression ratio (smokers/nonsmokers)
gene: HEMK1 | 218620_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1376
p=0.0311
p=0.0621
p=0.0898
p=0.5744
p=0.0971
p=0.9082
p=0.9265
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

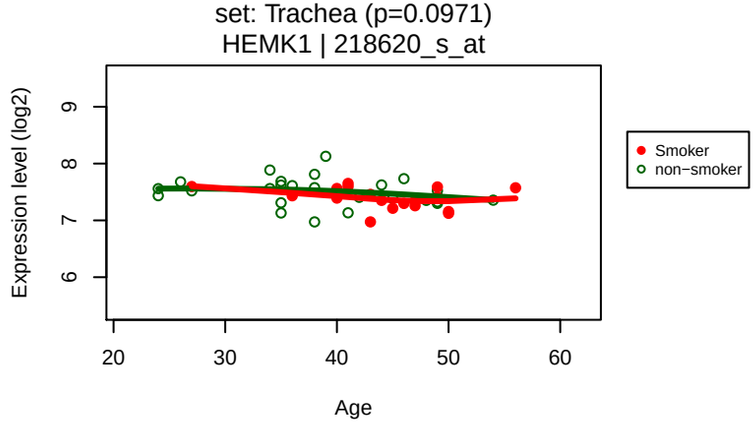

Gene expression ratio in Males
gene: HEMK1 | 218620_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.043
p=0.9026
p=0.1136
p=0.8512
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HEMK1 | 218620_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5672
p=0.1552
p=0.4966
p=0.9587
SmallAirway2
LargeAirway2
Trachea
Macrophage2

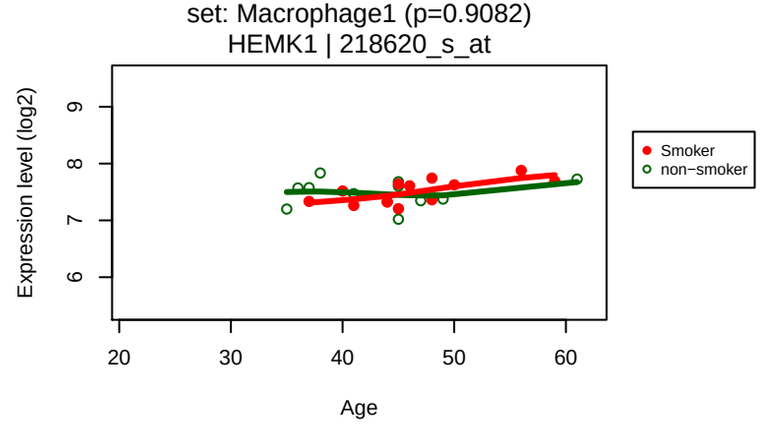

set: SmallAirway1 (p=0.1376)
HEMK1 | 218620_s_at
Expression level (log2)
Age
Smoker
non−smoker

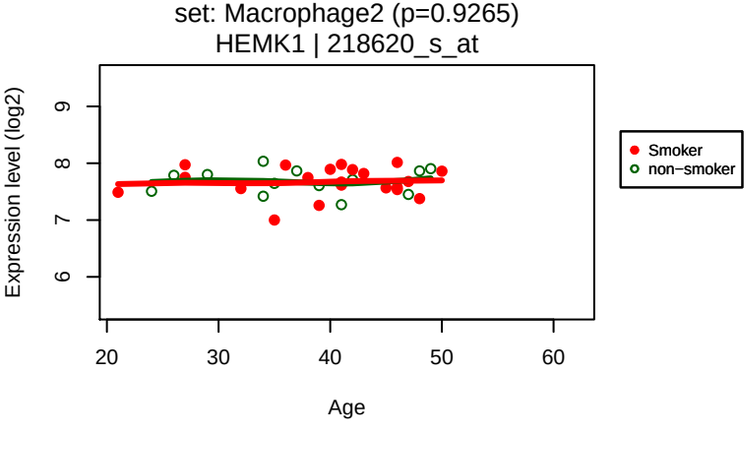

set: SmallAirway2 (p=0.0311)
HEMK1 | 218620_s_at
Expression level (log2)
Age
Smoker
non−smoker

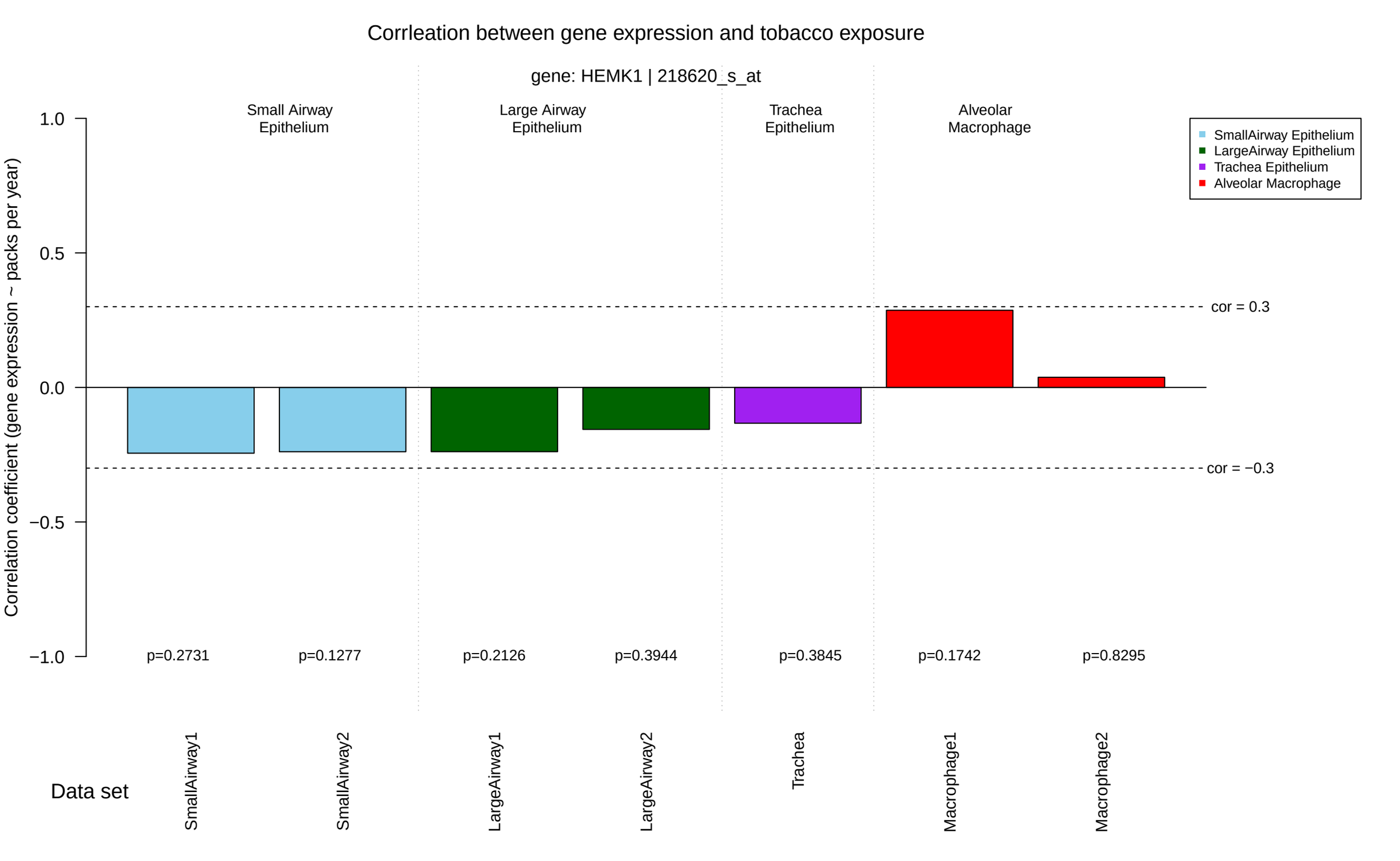

set: LargeAirway1 (p=0.0898)
HEMK1 | 218620_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5744)
HEMK1 | 218620_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0971)
HEMK1 | 218620_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9082)
HEMK1 | 218620_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9265)
HEMK1 | 218620_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HEMK1 | 218620_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2731
p=0.1277
p=0.2126
p=0.3944
p=0.3845
p=0.1742
p=0.8295
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2